Gas Problem Set: Ch 8 (8.1–8.5), 27 problems from easy to exam level

Work on scratch paper or in the spaces. Answers and full worked solutions are in *gas_problems_key.pdf*. ★ = exam-level problem. Use R = 0.08206 L·atm/(mol·K) unless told otherwise.

THE 5-STEP GAS ROUTINE (do this for EVERY gas problem)

1. **Make a table** of P, V, n, T. Write each given value with its unit and put "?" for the one you want. Two states (before/after)? Make two columns.
2. **Convert**: T → K (+273.15) · P → atm (÷760 for mmHg/Torr, ÷101.325 for kPa) · V → L (mL ÷ 1000) · grams → mol (÷$\mathcal{M}$).
3. **Pick the equation**:

Situation	Use
One state, find one of P, V, n, T	PV = nRT
Same gas, conditions change (before → after)	$P_1V_1/(n_1T_1) = P_2V_2/(n_2T_2)$; cross out what stays constant
Grams + volume given, find $\mathcal{M}$ or density	$\mathcal{M}$ = mRT/(PV) · d = P$\mathcal{M}$/(RT)
Mixture of gases	P_tot = ΣP_i · P_i = χ_i·P_tot · P_i = n_iRT/V
Collected over water	P_gas = P_total − P_{H_2O}
Reaction involving a gas	gas ⇄ mol via PV = nRT, then mole ratio. Same T and P? volume ratio = mole ratio
Rates / effusion	$rate_1/rate_2 = \sqrt{(\mathcal{M}_2/\mathcal{M}_1)}$ · $time_1/time_2 = \sqrt{(\mathcal{M}_1/\mathcal{M}_2)}$
Speeds / kinetic energy	u_rms = √(3RT/$\mathcal{M}$), R = 8.314, **$\mathcal{M}$ in kg/mol** · KE_avg = 3/2 RT (per mol)

4. **Predict first**: does the answer go up or down? (e.g. hotter → bigger V; more pressure → smaller V)
5. **Solve, then check** units, sig figs, and that the direction matches your prediction.

A. Pressure units & the simple gas laws

1. Convert 0.875 atm into mm-Hg, kPa, and bar. Then convert a pressure of 642 mm-Hg into atm.

2. (Boyle) A sealed syringe holds 50.0 mL of air at 1.00 atm. The plunger is pushed in until the volume is 12.5 mL at constant temperature. What is the new pressure?

3. (Charles) A 2.50-L balloon at 22.0 °C is dipped into liquid nitrogen (−196 °C) at constant pressure. What is its new volume?

4. (Gay-Lussac) A car tire is at 32.0 psi at 15 °C in the morning. After driving, the tire is at 45 °C. Assuming constant volume, what is the new pressure (treat psi as absolute pressure)?

5. ★ (Combined) A weather balloon holds 5.00 L of He at 755 mm-Hg and 25 °C at launch. It rises to an altitude where P = 380. mm-Hg and T = −15 °C. What is its volume there? Explain why it expanded even though it got colder.

6. (Avogadro) A 3.00-L container holds 0.125 mol of gas. If 0.0500 mol more gas is added at constant T and P, what will the new volume be?

B. Ideal gas law, PV = nRT

7. A 15.0-L scuba tank contains O_2 at 150. atm and 22 °C. How many moles and how many kilograms of O_2 does it hold?

8. 4.50 g of CO_2 is placed in a 2.00-L flask at body temperature (37 °C). What is the pressure in atm?

9. At what temperature (in K and °C) will 1.00 mol of a gas in a 22.4-L container exert a pressure of 2.00 atm?

C. Gas density & molar mass

10. Calculate the density of Cl_2 gas (70.90 g/mol) at STP and at 100. °C and 1.00 atm. Why does the density go down?

11. ★ A hydrocarbon contains 85.6% C and 14.4% H by mass. A 0.802-g sample of its vapor occupies 354 mL at 100. °C and 752 mm-Hg. Determine the molar mass and the molecular formula.

12. (Concept) At the same T and P, rank He, air ($\mathscr{M} \approx 29$ g/mol), and CO_2 by density. Which balloons float in air and which sink?

D. Dalton's law & collecting gas over water

13. ★ A 5.00-L flask at 25 °C contains 2.00 g He and 8.00 g O_2. Find the partial pressure of each gas, the total pressure, and the mole fraction of He.

14. Dry air is 20.95 mol-% O_2. On a day when the barometric pressure is 0.950 atm, what is the partial pressure of O_2 in atm and in mm-Hg?

15. ★ Heating $KClO_3$ gives O_2: $2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$. The O_2 is collected over water: 0.512 L at 22 °C and a total pressure of 758 mm-Hg (vapor pressure of water at 22 °C = 19.8 mm-Hg). What mass of $KClO_3$ (122.55 g/mol) decomposed?

E. Gas stoichiometry

16. ★ Car airbags inflate by $2\ NaN_3(s) \rightarrow 2\ Na(s) + 3\ N_2(g)$. What mass of NaN_3 (65.01 g/mol) is needed to fill a 65.0-L airbag at 1.10 atm and 25 °C?

17. 15.0 L H_2 and 6.00 L N_2 react at the same T and P: $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$. Which is limiting? What volume of NH_3 forms, and what volume of which gas is left over? (No ideal gas law needed!)

18. 10.0 g of propane (C_3H_8, 44.097 g/mol) burns: $C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$. (a) What volume of CO_2 forms at STP? (b) What volume of O_2 at 25 °C and 1.00 atm is consumed?

19. ★ 2.00 g of Al (26.982 g/mol) is reacted with 1.50 L of Cl_2 gas at 25 °C and 1.00 atm: $2\ Al(s) + 3\ Cl_2(g) \rightarrow 2\ AlCl_3(s)$. Determine the limiting reactant and the mass of $AlCl_3$ (133.33 g/mol) formed.

F. Effusion (Graham) & kinetic-molecular theory

20. (a) How many times faster does H_2 effuse than O_2? (b) An unknown gas effuses 0.354 times as fast as He. What is its molar mass, and what could it be? (c) 1.00 L of N_2 effuses through a pinhole in 105 s; the same volume of an unknown gas takes 191 s. Find the unknown's molar mass.

21. ★ Cotton balls soaked in $NH_3(aq)$ and HCl(aq) are placed at opposite ends of a 1.00-m glass tube at the same time. A white ring of $NH_4Cl(s)$ forms where the gases meet. How far from the NH_3 end does the ring form?

22. (a) Calculate u_rms for He and for Xe (131.29 g/mol) at 25 °C. (b) At what temperature would O_2 molecules have the same u_rms as He atoms at 25 °C?

23. Calculate the average kinetic energy of a mole of gas molecules and of a single molecule at 25 °C. Does it matter which gas?

24. (Concept) For a fixed amount of ideal gas, what happens to P, average KE, and u_rms when (a) T is doubled (in K) at constant V; (b) V is halved at constant T? Explain each with KMT.

G. Barometers & real gases

25. (a) How tall would a water barometer be at 1.00 atm ($\rho(Hg) = 13.56$ g/mL, $\rho(H_2O) = 1.00$ g/mL)? (b) What pressure (in atm) does a 10.0-m column of water exert ($P = \rho gh$, $g = 9.80665$ m/s²)?

26. (Concept) Which deviates more from ideal behavior and why: (a) He or NH_3 at 1 atm; (b) N_2 at 1 atm and 300 K or at 300 atm and 150 K? What are the two ideal-gas assumptions that fail, and what does Z = PV/(nRT) equal for an ideal gas?

H. Challenge

27. ★★ A rigid 2.00-L flask at 400. K contains 0.200 mol CO and 0.150 mol O_2. The reaction 2 CO(g) + O_2(g) → 2 CO_2(g) goes to completion at constant T. Find the total pressure before the reaction, the partial pressure of each gas after the reaction, and the total pressure after. Why did the pressure drop?